

**SIMATS ENGINEERING  
ASSESSMENT TOOL 1**

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**COURSE NAME: Cryptography and Network Security**

**COURSE CODE: CSA5112**

**APPLICATION- BASED QUESTIONS**

**1.Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message “MEET AT NOON” is transformed into an encoded version using a fixed shift pattern.**

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.**
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.**

**Answer**

Question 1: Scenario – Event Communication Security

1(a) Reconstruct the encrypted message using a shift of 4 positions

The Caesar Cipher is a classical encryption technique in which each letter of the plaintext is shifted by a fixed number of positions in the alphabet. Here, the shift value is 4, so every letter is moved four places forward.

Plaintext:

MEET AT NOON

Step-by-Step Encryption

| Plain Letter | Shift (+4) | Cipher Letter |
|--------------|------------|---------------|
| M            | Q          | Q             |
| E            | I          | I             |
| E            | I          | I             |
| T            | X          | X             |
| A            | E          | E             |
| T            | X          | X             |
| N            | R          | R             |
| O            | S          | S             |
| O            | S          | S             |
| N            | R          | R             |

Spaces remain unchanged.

Encrypted Message (Ciphertext):

QIIX EX RSSR

1(b) Explain how an authorized member retrieves the original message

To recover the original message, the receiver shifts each encrypted letter 4 positions backward because they know the encryption key.

#### Step-by-Step Decryption

| Cipher Letter | Shift (-4) | Plain Letter |
|---------------|------------|--------------|
| Q             | M          | M            |
| I             | E          | E            |
| I             | E          | E            |
| X             | T          | T            |
| E             | A          | A            |
| X             | T          | T            |
| R             | N          | N            |
| S             | O          | O            |
| S             | O          | O            |
| R             | N          | N            |

Recovered Plaintext:

MEET AT NOON

#### Conclusion

The Caesar Cipher encrypts a message by shifting each letter by a fixed number of positions. In this example, shifting each letter by 4 converts "MEET AT NOON" into "QIIX EX RSSR." The receiver can retrieve the original message by shifting each letter 4 positions backward. Although this cipher is easy to understand and implement, it is not secure for modern communication because it can be broken easily using brute-force or frequency analysis.

**2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.**

**a. Deduce the most likely original message.**

**b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.**

**A cybersecurity intern intercepts the coded message:**

**"GSRH RH Z HVXIVG"**

**The message appears to use a simple substitution cipher rather than a complex encryption algorithm. By analyzing the pattern of letters and words, it can be determined that the message is encrypted using the Atbash Cipher.**

2(a) Deduce the most likely original message

What is the Atbash Cipher?

The Atbash Cipher is one of the oldest classical substitution ciphers. Instead of shifting letters like the Caesar Cipher, it reverses the alphabet. The first letter is replaced with the last letter, the second with the second-last, and so on.

Atbash Mapping Table

| Plain | Cipher | Plain | Cipher |
|-------|--------|-------|--------|
| A     | Z      | N     | M      |
| B     | Y      | O     | L      |
| C     | X      | P     | K      |
| D     | W      | Q     | J      |
| E     | V      | R     | I      |
| F     | U      | S     | H      |
| G     | T      | T     | G      |
| H     | S      | U     | F      |
| I     | R      | V     | E      |
| J     | Q      | W     | D      |
| K     | P      | X     | C      |
| L     | O      | Y     | B      |
| M     | N      | Z     | A      |

Using this mapping, each letter of the ciphertext can be converted back into the original message. Step-by-Step Decryption

Ciphertext

GSRH RH Z HVXIVG

First Word: GSRH

| Cipher Letter | Plain Letter |
|---------------|--------------|
|---------------|--------------|

|   |   |
|---|---|
| G | T |
| S | H |
| R | I |
| H | S |

Result:

THIS

Second Word: RH

| Cipher Letter | Plain Letter |
|---------------|--------------|
|---------------|--------------|

|   |   |
|---|---|
| R | I |
| H | S |

Result:

IS

Third Word: Z

| Cipher Letter | Plain Letter |
|---------------|--------------|
|---------------|--------------|

|   |   |
|---|---|
| Z | A |
|---|---|

Result:

A

Fourth Word: HVXIVG

| Cipher Letter | Plain Letter |
|---------------|--------------|
|---------------|--------------|

|   |   |
|---|---|
| H | S |
| V | E |
| X | C |
| I | R |
| V | E |
| G | T |

Result:  
SECRET

Recovered Plaintext  
THIS IS A SECRET

Hence, the most likely original message is:  
THIS IS A SECRET

2(b) Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics

The intercepted message can be identified as an Atbash Cipher by analyzing several characteristics.

#### 1. Word Length Analysis

The encrypted message contains four words:

- GSRH (4 letters)
- RH (2 letters)
- Z (1 letter)
- HVXIVG (6 letters)

After decryption, the word lengths remain exactly the same:

- THIS (4 letters)
- IS (2 letters)
- A (1 letter)
- SECRET (6 letters)

Since substitution ciphers replace letters without changing the length of words, this strongly supports the Atbash Cipher.

#### 2. Repeating Letter Pattern

In the encrypted word HVXIVG, the letter V appears twice.

After decryption:

HVXIVG → SECRET

The repeated V becomes the repeated E in the word "SECRET."

This preservation of repeated letters is a common feature of substitution ciphers.

#### 3. Symmetric Alphabet Mapping

Unlike the Caesar Cipher, the Atbash Cipher uses a reversed alphabet.

Examples include:

- A ↔ Z
- B ↔ Y
- C ↔ X
- D ↔ W
- E ↔ V

Similarly,

- G ↔ T
- H ↔ S
- R ↔ I

Applying this reverse mapping to every letter immediately produces meaningful English words.

#### 4. Common English Phrase

Recognition After decoding, the message becomes:

THIS IS A SECRET

This is a meaningful and grammatically correct English sentence. Since intercepted

communications often contain simple instructions or confidential statements this phrase is highly believable.

#### 5. Atbash Cipher is Self-Inverse

One special property of the Atbash Cipher is that the same substitution table is used for both encryption and decryption.

For example:

- $G \rightarrow T$
- $T \rightarrow G$
- $H \rightarrow S$
- $S \rightarrow H$

Applying the Atbash substitution twice returns the original message. This property makes decryption simple once the cipher type is recognized.

#### 6. Difference Between Caesar Cipher and Atbash Cipher

| Caesar Cipher                                                                                                        | Atbash Cipher                                  |
|----------------------------------------------------------------------------------------------------------------------|------------------------------------------------|
| Uses a fixed shift value                                                                                             | Uses a reversed alphabet                       |
| Requires a key                                                                                                       | No key is required                             |
| Different keys produce different ciphertexts                                                                         | Same mapping is always used                    |
| Encryption and decryption use opposite shifts                                                                        | Encryption and decryption use the same mapping |
| The intercepted message does not match any simple Caesar shift, but it is perfectly decoded using the Atbash Cipher. |                                                |

#### Advantages of Atbash Cipher

- Very easy to understand and implement.
- No encryption key is required.
- Encryption and decryption use the same method.
- Useful for learning substitution ciphers.
- Historically significant as one of the oldest known ciphers.

#### Limitations of Atbash Cipher

- Provides very weak security.
- The substitution pattern is fixed and publicly known.
- Can be broken easily using frequency analysis.
- Not suitable for protecting sensitive information in modern communication.

#### Applications of Atbash Cipher

Although it is not used for real-world security today, the Atbash Cipher is still useful in:

- Cryptography education.
- Puzzle solving and brain teasers.
- Historical studies of encryption.
- Coding competitions and cybersecurity practice.
- Demonstrating substitution cipher concepts.

#### Conclusion

The intercepted message "GSRH RH Z HVXIVG" was encrypted using the Atbash Cipher, a classical substitution cipher that reverses the alphabet. By replacing each letter with its

opposite letter in the alphabet, the original message is successfully recovered as "THIS IS A SECRET." The repeated letter patterns, unchanged word lengths, and reversed alphabet mapping clearly indicate the use of the Atbash Cipher. While the cipher is simple and useful for educational purposes, it offers very little security and is unsuitable for protecting confidential information in modern communication system.